Roll No. 124104069

EEPC204: Signals and Systems

Dec. 2025

Time: 3 Hrs. *MM:50*

National Institute of Technology
Department of Electrical Engineering
THEORY EXAMINATION (REGULAR)

Programme: B.Tech
Total No. of Questions: 6

Semester: 3rd

No. of Pages: 2
No. of Questions to be Attempted: 5

Instructions:

1. Attempt all subparts of a question together.
2. Unless stated otherwise, the symbols have their usual meanings in context with the subject.
3. Assume suitably and state any additional data required, if any.
4. Before starting, check the question paper for discrepancies and ensure you have the correct course and subject.

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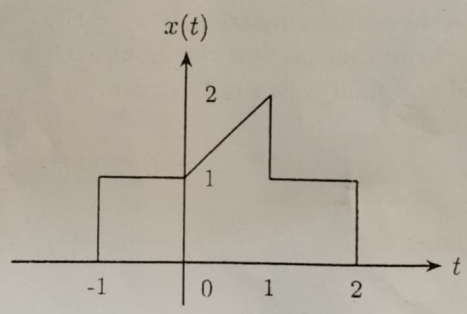
1. a) For the signal $x(t)$ shown in Figure, sketch and label $x(t+3)$, $x\left(\frac{1}{2}t-2\right)$ and $x(-t-2)$. [3]



b) Explain what is meant by a system. Define the following key properties of a system: [4]
   i) Linear and Nonlinear
   ii) Time-Varying and Time-Invariant (TIV)
   iii) BIBO Stability

c) Find whether the signal

$$x(t) = \begin{cases} t-2, & -2 \le t \le 0 \\ 2-t, & 0 \le t \le 2 \\ 0, & \text{otherwise} \end{cases}$$

is an energy signal or a power signal. Also, find the energy and power of the signal. [3]

2. a) Find the convolution of the following signals using the graphical method: [6]

$$x(t) = e^{-3t}\,u(t), \quad h(t) = u(t+3)$$

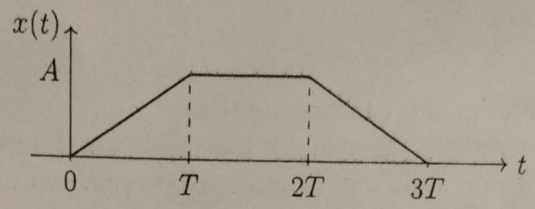
b) A student at NIT Kurukshetra is working on designing a finite-duration discrete-time filter. Accordingly, he designed a filter whose impulse response is given by $h[n] = u[n] - u[n-3]$. The student then applied a finite length test signal $x[n] = \{1, 1, 0, 1\}$ to this filter. Determine the output signal he observed. Also, explain how the filter would behave if the input signal were infinite in length. [4 marks]

+2

3. a) Find the exponential Fourier series of the periodic signal with period $T = 4$. One period (centered at $t = 0$) is [5]

$$x(t) = \begin{cases} 0, & |t| \geq 2, \\ t+2, & -2 \leq t < -1, \\ 1, & -1 \leq t \leq 1, \\ 2-t, & 1 < t \leq 2. \end{cases}$$

b) Find the Fourier transform of the signal $x(t)$ shown in the figure below. [5]



4. a) Consider the continuous-time LTI system: [3+3]

$$\frac{d^3y(t)}{dt^3} - 4\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} - 2y(t) = \frac{dx(t)}{dt} + 3x(t).$$

(i) Determine the transfer function $H(s) = Y(s)/X(s)$.
(ii) Compute the output $y(t)$ for the input $x(t) = e^{-2t}u(t)$.

b) Find the initial and final values of the following: [2]

$$G(s) = \frac{s+3}{s^2+5s+4}$$

c) Evaluate the integral: $I = \int_{-3}^{2} \left(e^{-t+1} + \sin\left(\frac{2\pi t}{3}\right)\right) \delta\left(t - \frac{3}{2}\right) dt$, where $\delta(t)$ is the Dirac delta function. [2]

5. (a) Prove that convolution in time domain is equal to multiplication in frequency domain. [3]
(b) State sampling theorem. Prove the process of recovery of the original signal from its sampled version. Also explain the significance of sampling theorem. Determine the Nyquist rate for a continuous-time signal $f(t) = 6\cos 50\pi t + 20\sin 300\pi t + 10\cos 100\pi t$. [7]

6. (a) Determine inverse z-transform $h(n)$ for the given [3]

$$H(z) = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}.$$

Also comment on the stability.

(b) Obtain impulse response of a causal LTI system [3]

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n].$$

(c) For a signal $x[n]$ with z-transform $X(z)$, the following five facts are given: [4]

i. $x[n]$ is real and right-sided,
ii. $X(z)$ has exactly two poles,
iii. $X(z)$ has two zeros at the origin,
iv. $X(z)$ has a pole at $z = \frac{1}{2}e^{j\pi/3}$,
v. $X(1) = \frac{8}{3}$.

Determine $X(z)$ and specify its ROC.